

SM PHARMACEUTICALS SDN BHD

PREGAL CAPSULES
(Pregabalin 75 mg and 150 mg)

DESCRIPTION:
75 mg: Size '4' Capsules with dark brown cap and white body imprinted with "SM-LU" on cap and "75" on body in black ink containing white to off white powder.
150 mg: Size '2' Capsules with white cap and white body imprinted with "SM-LU" on cap and "150" on body in black ink containing white to off white powder.

COMPOSITION:
75 mg: Each capsule contains: Pregabalin 75mg
150mg: Each capsule contains: Pregabalin 150mg

PHARMACODYNAMICS:
The active substance, pregabalin, is a gamma=aminobutyric acid analogue [(5)-3-(aminomethyl)-5-methylhexanoic acid].
Mechanism of action:
Pregabalin binds to an auxiliary subunit (α-δ protein) of voltage-gated calcium channels in the central nervous system.
Clinical efficacy and safety
Neuropathic pain
Efficacy has been shown in diabetic neuropathy, post herpetic neuralgia and spinal cord injury. Efficacy has not been studied in other models of neuropathic pain.
Epilepsy
Adjunctive Treatment
Pregabalin has been studied with either BID or TID dosing. Overall, the safety and efficacy profiles for BID and TID dosing regimens are similar. A reduction in seizure frequency was observed by Week 1.
Paediatric population
The efficacy and safety of pregabalin as adjunctive treatment for epilepsy in paediatric patients below the age of 12 and adolescents has not been established.
Monotherapy (newly diagnosed patients)
Pregabalin and lamotrigine were similarly safe and well tolerated.
Generalised Anxiety Disorder
Relief of the symptoms of GAD as reflected by the Hamilton Anxiety Rating Scale (HAM-A) has been observed.

PHARMACOKINETICS:
Pregabalin steady-state pharmacokinetics are similar in healthy volunteers, patients with epilepsy receiving anti-epileptic drugs and patients with chronic pain.
Absorption
Pregabalin is rapidly absorbed when administered in the fasted state, with peak plasma concentrations occurring within 1 hour following both single and multiple dose administration. Pregabalin oral bioavailability is estimated to be ≥ 90% and is independent of dose. Following repeated administration, steady state is achieved within 24 to 48 hours. The rate of pregabalin absorption is decreased when given with food resulting in a decrease in C_{max} by approximately 25-30% and a delay in t_{max} to approximately 2.5 hours. However, administration of pregabalin with food has no clinically significant effect on the extent of pregabalin absorption.
Distribution:
In humans, the apparent volume of distribution of pregabalin following oral administration is approximately 0.56 l/kg. Pregabalin is not bound to plasma proteins.
Biotransformation:
Pregabalin undergoes negligible metabolism in humans.
Elimination:
Pregabalin is eliminated from the systemic circulation primarily by renal excretion as unchanged drug. Pregabalin mean elimination half-life is 6.3 hrs. Pregabalin plasma clearance and renal clearance are directly proportional to creatinine clearance. Dose adjustment in patients with reduced renal function or undergoing haemodialysis is necessary.
Linearity / non-linearity
Pregabalin pharmacokinetics are linear over the recommended daily dose range. Multiple dose pharmacokinetics are predictable from single-dose data. Therefore, there is no need for routine monitoring of plasma concentrations of pregabalin.
Gender:
Gender does not have a clinically significant influence on the plasma concentrations of pregabalin.
Renal Impairment:
Pregabalin clearance is directly proportional to creatinine clearance. In addition, pregabalin is effectively removed from plasma by haemodialysis (following a 4 hour haemodialysis treatment plasma pregabalin concentrations are reduced by approximately 50%). Because renal elimination is the major elimination pathway, dose reduction in patients with renal impairment and dose supplementation following haemodialysis is necessary.
Hepatic Impairment:
Since pregabalin does not undergo significant metabolism and is excreted predominantly as unchanged drug in the urine, impaired liver function would not be expected to significantly alter pregabalin plasma concentrations.
Paediatric population
After oral administration of pregabalin in pediatric patients in the fasted state, in general, time to reach peak plasma concentration was similar across the entire age group and occurred 0.5 hours to 2 hours post dose.
Pregabalin C_{max} and AUC parameters increased in a linear manner with increasing dose within each age group. The AUC was lower by 30% in paediatric patients below a weight of 30 kg due to an increased body weight adjusted clearance of 43% for these patients in comparison to patients weighing ≥30 kg. Pregabalin terminal half-life averaged about 3 to 4 hours in paediatric patients up to 6 years of age, and 4 to 6 hours in those 7 years of age and older.
Population pharmacokinetic analysis showed that creatinine clearance was a significant covariate of pregabalin oral clearance, body weight was a significant covariate of pregabalin apparent oral volume of distribution, and these relationships were similar in paediatric and adult patients.
Elderly
Pregabalin clearance tends to decrease with increasing age. This decrease in pregabalin oral clearance is consistent with decreases in creatinine clearance associated with increasing age. Reduction of pregabalin dose may be required in patients who have age-related compromised renal function.
Breast-feeding mothers
Lactation had little to no influence on pregabalin pharmacokinetics.
Pregabalin was excreted into breast milk with average steady-state concentrations approximately 76% of those in maternal plasma. The estimated infant dose from breast milk (assuming mean milk consumption of 150 ml/kg/day) of women receiving 300 mg/day or the maximum dose of 600 mg/day would be 0.31 or 0.62 mg/kg/day, respectively. These estimated doses are approximately 7% of the total daily maternal dose on a mg/kg basis.

INDICATIONS:
Neuropathic pain
Pregabalin is indicated for the treatment of peripheral and central neuropathic pain in adults.
Epilepsy
Pregabalin is indicated as adjunctive therapy in adults with partial seizures with or without secondary generalisation.
Generalised Anxiety Disorder
Pregabalin is indicated for the treatment of Generalised Anxiety Disorder (GAD) in adults.
Fibromyalgia
Pregabalin is indicated for the management of fibromyalgia.

CONTRAINDICATIONS:
Hypersensitivity to the active substance or to any of the excipients.

SIDE EFFECTS / ADVERSE REACTIONS:
In table 2 below all adverse reactions, which occurred at an incidence greater than placebo and in more than one patient are listed by class and frequency.
Within each frequency grouping, undesirable effects are presented in order of decreasing seriousness. The adverse reactions listed may also be associated with the underlying disease and/or concomitant medicinal products.
In the treatment of central neuropathic pain due to spinal cord injury the incidence of adverse reactions in general, CNS adverse reactions and especially somnolence was increased.
Additional reactions reported from postmarketing experience are included in italics in the list below.

Table 2. Pregabalin Adverse Drug Reactions	
System	Organ
Adverse Drug Reactions	
Class	
Infections and infestations	
Common	Nasopharyngitis
Blood and lymphatic system disorders	
Uncommon	Neutropenia
Immune system disorders	
Uncommon	<i>Hypersensitivity</i>
Rare	<i>Angioedema, allergic reaction</i>
Metabolism and nutrition disorders	
Uncommon	Appetite increased
Common	Anorexia, hypoglycaemia
Psychiatric disorders	
Common	Euphoric mood, confusion, irritability, disorientation, insomnia, libido decreased
Uncommon	Hallucination, panic attack, restlessness, agitation, depression, depressed mood, elevated mood, <i>aggression</i> , mood swings, depersonalisation, word finding difficulty, abnormal dreams, libido increased, <i>anorgasmia, apathy</i>
Rare	Disinhibition
Nervous system disorders	
Very Common	Dizziness, somnolence, headache
Common	Ataxia, coordination abnormal, tremor, dysarthria, amnesia, memory impairment, disturbance in attention, paraesthesia, hypoaesthesia, sedation, balance disorder, lethargy
Uncommon	Syncope, stupor, myoclonus, <i>loss of consciousness</i> , psychomotor hyperactivity, dyskinesia, dizziness postural, intention tremor, nystagmus, cognitive disorder, <i>mental impairment</i> , speech disorder, hyporeflexia, hyperaesthesia, burning sensation, <i>ageusia, malaise</i>
Rare	<i>Convulsions, parosmia, hypokinesia, dysgraphia</i>
Eye disorders	
Common	Vision blurred, diplopia
Uncommon	Peripheral vision loss, Visual disturbance, eye swelling, visual field defect, visual acuity reduced, eye pain, asthenopia, photopsia, dry eye, lacrimation increased, eye irritation
Rare	Vision loss, keratitis, oscillopsia, altered vision depth perception, mydriasis, strabismus, visual brightness
Ear and labyrinth disorders	
Common	Vertigo
Uncommon	Hyperacusis
Cardiac disorders	
Uncommon	Tachycardia, atrioventricular block first degree, sinus bradycardia, <i>congestive heart failure</i>
Rare	<i>QT prolongation</i> , sinus tachycardia, sinus arrhythmia
Vascular disorders	
Uncommon	Hypotension, hypertension, hot flushes, flushing, peripheral coldness
Respiratory, thoracic and mediastinal disorders	
Uncommon	Dyspnoea, epistaxis, cough, nasal congestion, rhinitis, snoring, nasal dryness
Rare	<i>Pulmonary oedema</i> , throat tightness
Not known	Respiratory depression
Gastrointestinal disorders	
Common	Vomiting, <i>nausea</i> , constipation, <i>diarrhoea</i> , flatulence, abdominal distension, dry mouth
Uncommon	Gastrooesophageal reflux disease, salivary hypersecretion, hypoaesthesia

	oral
Rare	Ascites, pancreatitis, <i>swollen tongue</i> , dysphagia
Hepatobiliary disorders	
Uncommon	Elevated liver enzymes*
Rare	Jaundice
Very rare	Hepatic failure, hepatitis
Skin and subcutaneous tissue disorders	
Uncommon	Rash papular, urticaria, hyperhidrosis, <i>pruritus</i>
Rare	<i>Sievens Johnson syndrome</i> , cold sweat
Musculoskeletal and connective tissue disorders	
Common	Muscle cramp, arthralgia, back pain, pain in limb, cervical spasm
Uncommon	Joint swelling, myalgia, muscle twitching, neck pain, muscle stiffness
Rare	Rhabdomyolysis
Renal and urinary disorders	
Uncommon	Urinary incontinence, dysuria
Rare	Renal failure, oliguria, <i>urinary retention</i>
Reproductive system and breast disorders	
Common	Erectile dysfunction
Uncommon	Ejaculation delayed, <i>sexual dysfunction</i> , <i>dysmenorrhoea</i> , <i>breast pain</i>
Rare	<i>Amenorrhoea</i> , breast discharge, breast enlargement, <i>gynaecomastia</i>
General disorders and administration site conditions	
Common	Gait abnormal, feeling drunk, fatigue, oedema peripheral, oedema, fall, feeling abnormal
Uncommon	Generalised oedema, pyrexia, <i>face oedema</i> , chest tightness, pain, thirst, chills, asthenia
Investigations	
Common	Weight increased
Uncommon	Blood creatine phosphokinase increased, alanine aminotransferase increased, aspartate aminotransferase increased, blood glucose increased, platelet count decreased, blood creatinine increased, blood potassium decreased, weight decreased.
Rare	White blood cell count decreased

* Alanine aminotransferase increased (ALT) and aspartate aminotransferase increased (AST).
After discontinuation of short-term and long-term treatment with pregabalin withdrawal symptoms have been observed in some patients. The following reactions have been mentioned: insomnia, headache, nausea, anxiety, diarrhoea, flu syndrome, convulsions, nervousness, depression, pain, hyperhidrosis and dizziness, suggestive of physical dependence. The patient should be informed about this at the start of the treatment.
Concerning discontinuation of long-term treatment of pregabalin, data suggest that the incidence and severity of withdrawal symptoms may be dose-related.
Paediatric population
The pregabalin safety profile observed was similar to that observed in the adult studies of patients with epilepsy. The most common adverse events observed were somnolence, pyrexia, upper respiratory tract infection, increased appetite, weight increased, and nasopharyngitis.

PRECAUTIONS / WARNINGS:
Diabetic patients
Diabetic patients who gain weight on pregabalin treatment may need to adjust hypoglycemic medicinal products.
Hypersensitivity reactions
There have been reports of hypersensitivity reactions, including cases of angioedema. Pregabalin should be discontinued immediately if symptoms of angioedema, such as facial, perioral, or upper airway swelling occur.
Dizziness, somnolence, loss of consciousness, confusion and mental impairment
Pregabalin treatment has been associated with dizziness and somnolence, which could increase the occurrence of accidental injury (fall) in the elderly population. There have also been reports of loss of consciousness, confusion and mental impairment. Therefore, patients should be advised to exercise caution until they are familiar with the potential effects of the medicinal product.
Vision-related effects
visual adverse reactions, including loss of vision, visual blurring or other changes of visual acuity, many of which were transient. Discontinuation of pregabalin may result in resolution or improvement of these visual symptoms.

Renal Failure
Cases of renal failure have been reported and in some cases discontinuation of pregabalin did show reversibility of this adverse reaction.

Withdrawal symptoms
After discontinuation of short-term and long-term treatment with pregabalin, withdrawal symptoms have been observed which include insomnia, headache, nausea, anxiety, diarrhoea, flu syndrome, nervousness, depression, pain, convulsion, hyperhidrosis and dizziness, suggestive of physical dependence. The patient should be informed about this at the start of the treatment.
Convulsions, including status epilepticus and grand mal convulsions, may occur during pregabalin use or shortly after discontinuing pregabalin.
Concerning discontinuation of long-term treatment of pregabalin, the incidence and severity of withdrawal symptoms may be dose-related.
Congestive heart failure
Congestive heart failure are mostly seen in elderly cardiovascular compromised patients during pregabalin treatment for a neuropathic indication. Pregabalin should be used with caution in these patients. Discontinuation of pregabalin may resolve the reaction.

Treatment of central neuropathic pain due to spinal cord injury
In the treatment of central neuropathic pain due to spinal cord injury the incidence of adverse reactions in general, central nervous system adverse reactions and especially somnolence was increased. This may be attributed to an additive effect due to concomitant medicinal products (e.g. anti-spasticity agents) needed for this condition. This should be considered when prescribing pregabalin in this condition.
Respiratory depression
There have been reports of severe respiratory depression in relation to pregabalin use. Patients with compromised respiratory function, respiratory or neurological disease, renal impairment, concomitant use of CNS depressants and the elderly may be at higher risk of experiencing this severe adverse reaction. Dose adjustments may be necessary in these patients.

Suicidal ideation and behavior
Suicidal ideation and behavior have been reported in patients treated with anti-epileptic agents in several indications.
Patients should be monitored for signs of suicidal ideation and behaviors and appropriate treatment should be considered. Patients (and caregivers of patients) should be advised to seek medical advice should signs of suicidal ideation or behavior emerge.
Reduced lower gastrointestinal tract function
Reduced lower gastrointestinal tract function (e.g. intestinal obstruction, paralytic ileus, constipation) have been reported when pregabalin is co-administered with medications that have the potential to produce constipation, such as opioid analgesics. When pregabalin and opioids are used in combination, measures to prevent constipation may be considered (especially in female patients and elderly).
Concomitant use with opioids
Caution is advised when prescribing pregabalin concomitantly with opioids due to risk of CNS depression.
Misuse, abuse potential or dependence
Caution should be exercised in patients with a history of substance abuse and the patient should be monitored for symptoms of pregabalin misuse, abuse or dependence (development of tolerance, dose escalation, drug-seeking behavior have been reported).
Encephalopathy
Cases of encephalopathy have been reported, mostly in patients with underlying conditions that may precipitate encephalopathy.
Lactose intolerance
Pregal contains lactose monohydrate. Patients with rare hereditary problems of galactose intolerance, the Lapp lactase deficiency or glucose-galactose malabsorption should not take this medicinal product.
Sodium content
Pregal contains less than 1 mmol sodium (23 mg) per hard capsule. Patients on low sodium diets can be informed that this medicinal product is essentially 'sodium-free'.

WARNING AND PECAUTIONS
Potential for an increase in risk of suicidal thoughts or behaviors.

EFFECTS ON ABILITY TO DRIVE AND USE MACHINE:
Pregabalin may have minor or moderate influence on the ability to drive and use machines. Pregabalin may cause dizziness and somnolence and therefore may influence the ability to drive or use machines. Patients are advised not to drive, operate complex machinery or engage in other potentially hazardous activities until it is known whether this medicinal product affects their ability to perform these activities

DRUG INTERACTIONS:
Since Pregabalin is predominantly excreted unchanged in the urine, undergoes negligible metabolism in humans and is not bound to plasma proteins, it is unlikely to produce, or be subject to, pharmacokinetic interactions.
Accordingly, there is no clinically relevant pharmacokinetic interactions –between pregabalin and phenytoin, carbamazepine, valproic acid, lamotrigine, gabapentin, lorazepam, oxycodone or ethanol. Population pharmacokinetic analysis indicated that oral antidiabetics, diuretics, insulin, phenobarbital, tiagabine and topiramate had no clinically significant effect on pregabalin clearance.

Oral contraceptives, norethisterone and/or ethinyl oestradiol
Co-administration of pregabalin with the oral contraceptives norethisterone and/or ethinyl oestradiol does not influence the steady-state pharmacokinetics of either substance.

Central nervous system influencing medical products
Pregabalin may potentiate the effects of ethanol and lorazepam.

There are reports of respiratory failure, coma and deaths in patients taking pregabalin and opioids and/or other central nervous system (CNS) depressant medicinal products. Pregabalin appears to be additive in the impairment of cognitive and gross motor function caused by oxycodone.

PREGNANCY AND LACTATION
Women of childbearing potential / Contraception in males and females
As the potential risk for humans is unknown, effective contraception must be used in women of child bearing potential.
Pregnancy
There are no adequate data from the use of pregabalin in pregnant women.
Pregabalin should not be used during pregnancy unless clearly necessary (if the benefit to the mother clearly outweighs the potential risk to the foetus).
Breast-feeding
Pregabalin is excreted into human milk. The effect of Pregabalin on new-born/infants is unknown. A decision must be made whether to discontinue breast-feeding or to discontinue Pregabalin therapy taking into account the benefit of breast-feeding for the child and the benefit of therapy for the woman.
Fertility
There are no clinical data on the effects of pregabalin on female fertility.

RECOMMENDED DOSAGE, DOSAGE SCHEDULE AND ROUTE OF ADMINISTRATION:
The dose range is 150 to 600 mg per day given in either two or three divided doses.
Pregabalin may be taken with or without food.
Neuropathic pain
Pregabalin treatment can be started at a dose of 150 mg per day. Based on individual patient response and tolerability, the dosage may be increased to 300 mg per day after

an interval of 3 to 7 days, and if needed, to a maximum dose of 600 mg per day after an additional 7-day interval.

Fibromyalgia
The usual dose range for most patients is 300 to 450 mg per day given in two divided doses. Some patients may derive additional benefit at 600 mg per day. Dosing should begin at 75 mg two times a day (150 mg/day) and may be increased to 150 mg two times a day (300 mg/day) within 1 week based on efficacy and tolerability. Patients who do not experience sufficient benefit with 300 mg/day may be further increased to 225 mg two times a day (450 mg/day). If needed, in some patients, based on individual response and tolerability, the dose may be increased to maximum dosage of 600 mg/day after an additional week.

Epilepsy
Pregabalin treatment can be started with a dose of 150 mg per day. Based on individual patient response and tolerability, the dosage may be increased to 300 mg per day after 1 week. The maximum dosage of 600 mg per day may be achieved after an additional week.

*** Generalised Anxiety Disorder:**
The dose range is 150 to 600 mg per day given as two or three divided doses. The need for treatment should be reassessed regularly.

Pregabalin treatment can be started with a dose of 150 mg per day. Based on individual patient response and tolerability, the dosage may be increased to 300 mg per day after 1 week. Following an additional week, the dose may be increased to 450 mg per day. The maximum dosage of 600 mg per day may be achieved after an additional week.

Discontinuation of Pregabalin:
If pregabalin has to be discontinued, it is recommended this should be done gradually over a minimum of 1 week.

Patients with Renal impairment:
Dosage reduction in patients with compromised renal function must be individualised according to creatinine clearance (CL_{cr}), as indicated in Table 1 determined using the following formula:

$$CL_{cr}(mL/min) = \left(\frac{[140 - \text{age (years)}] \times \text{weight (kg)}}{72 \times \text{serum creatinine (mg/dL)}} \right) \quad (\times 0.85 \text{ for female patients})$$

For patients receiving haemodialysis, the pregabalin daily dose should be adjusted based on renal function. In addition to the daily dose, a supplementary dose should be given immediately following every 4 hour haemodialysis treatment (see Table 1).

Table 1. Pregabalin Dosage Adjustment Based on Renal Function			
		Total pregabalin daily dose*	
Creatinine clearance (CL _{cr}) (mL/min)	Starting dose (mg/day)	Maximum dose (mg/day)	Dose regimen
≥ 60	150	600	BID or TID
≥ 30 - < 60	75	300	BID or TID
≥ 15 - < 30	25 – 50	150	QD or BID
< 15	25	75	QD
Supplementary dosage following haemodialysis (mg)			
	25	100	Single dose +

TID = Three divided doses
BID = Two divided doses
QD = Single daily dose
* Total daily dose (mg/day) should be divided as indicated by dose regimen to provide mg/dose
* Supplementary dose is a single additional dose.
Use in patients with hepatic impairment
No dosage adjustment is required for patients with hepatic impairment.
Use in children and adolescents (12 to 17 years of age)
The safety and effectiveness of pregabalin in pediatric patients below the age of 12 years and adolescents has not been established.
The use in children is not recommended.
Use in elderly (over 65 years of age)
Elderly patients may require a dose reduction of pregabalin due to a decreased renal function.

Route of Administration: Oral (Solid Dosage Form)

SYMPTOMS AND TREATMENT FOR OVERDOSE AND ANTIDOTE(S):
Overdose symptoms included somnolence, confusion state, agitation, restlessness and seizures. Treatment of Pregabalin overdose should include general supportive measures and may include haemodialysis if necessary.

PACKING / PACK SIZES:
8 x 7's (56 Capsules).

STORAGE CONDITIONS, USER INSTRUCTIONS AND PHARMACEUTICAL PRECAUTIONS:
Store below 30°C. Keep in a dry place away from sunlight.
Instructions to Use:
Refer Recommended Dosage, Dosage Schedule and Route of Administration
The dosage range is 150 to 600 mg per day given in either two or three divided doses and may be taken with or without food.
If pregabalin has to be discontinued, it is recommended this should be done gradually over a minimum of 1 week independent of the indication.

SHELF LIFE:
3 years

LIST OF EXCIPIENTS

Pregelatinized starch

Talc

NAME AND ADDRESS OF MANUFACTURER:
SM PHARMACEUTICALS SDN BHD (218620-M)
Lot 88, Sungai Petani Industrial Estate
08000 Sungai Petani
Kedah Darul Aman, Malaysia.

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